

Ernest Choi

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Projects

AI Resume & Job-Search Automation Bot | *Python, LLM APIs, SQLite* github.com/ErChoi-com/DiscordBot

- Developed a **Python** bot that gathers and verifies job posting information through **REST APIs** and browser automation for entry into a computer system, processing 206,775 postings.
- Designed an **LLM-driven** pipeline to generate structured **JSON** output.

Experience

Technical Development Intern, City of Markham – Markham, Ontario July 2022 – August 2024

- Maintained clear, accurate documentation of **CITY** website workflows and operating processes, helping cut website downtime by 41%.
- Tracked project milestones and reported progress to management using Microsoft Office tools like **Excel** and **Word**, reducing coordination delays by 14%.
- Validated and tested system changes against quality standards and functional requirements, reducing reported errors by 20%.

Engineering Instructor, Obotz Robotics – Markham, Ontario September 2022 – Present

- Explained students' technical progress to parents in clear, non-technical terms.
- Taught **technical** concepts and embedded **C** programming, raising students' robotics problem-completion scores by 35%.
- Coached students to place in the 75th percentile of robotics competitions, requiring strong organizational skills and attention to detail.

Machine Learning Intern, Goppter Commerce Solutions – Remote September 2025 – December 2025

- Engineered a low-latency data pipeline for financial modeling, streaming **Kafka** data into **Python** services to ensure accurate mathematical calculations.
- Deployed analytics dashboards in **Tableau** and **Power BI**, surfacing real-time predictive information and detailed insights.
- Built reinforcement learning models in **TensorFlow** to generate adaptive trading signals.

Frontend Development Intern, MCG3D – Remote January 2026 – March 2026

- Wrote **technical** documentation and deployed applications in **Docker** containers.
- Built production web applications in **React** and **TypeScript**, maintaining a user-centric and responsive interface.
- Integrated frontend features with **Python Flask** backend services over secure **RESTful APIs**.

Skills

Languages: JavaScript/Node.js, Java, Python, C++, C, SQL, HTML, CSS

Frameworks/Libraries: OpenGL, WebRTC, JavaFX, React, TypeScript, Flask, FastAPI, TensorFlow

Tools: Microsoft Suite, Git, PostgreSQL, Ubuntu Linux, Docker, Kafka, Tableau, Power BI

Platforms: GitHub, GitLab, Raspberry Pi, Arduino

Relevant Courses: Object Oriented Eng Analysis and Design, Digital Systems, Electronic Circuits I, Microprocessor Systems

Education

Toronto Metropolitan University – Bachelor of Engineering in Computer Engineering 2023–2027